MASTER'S THESIS · PROGRESS UPDATE 1

Automatic Detection of Sleep Graphoelements

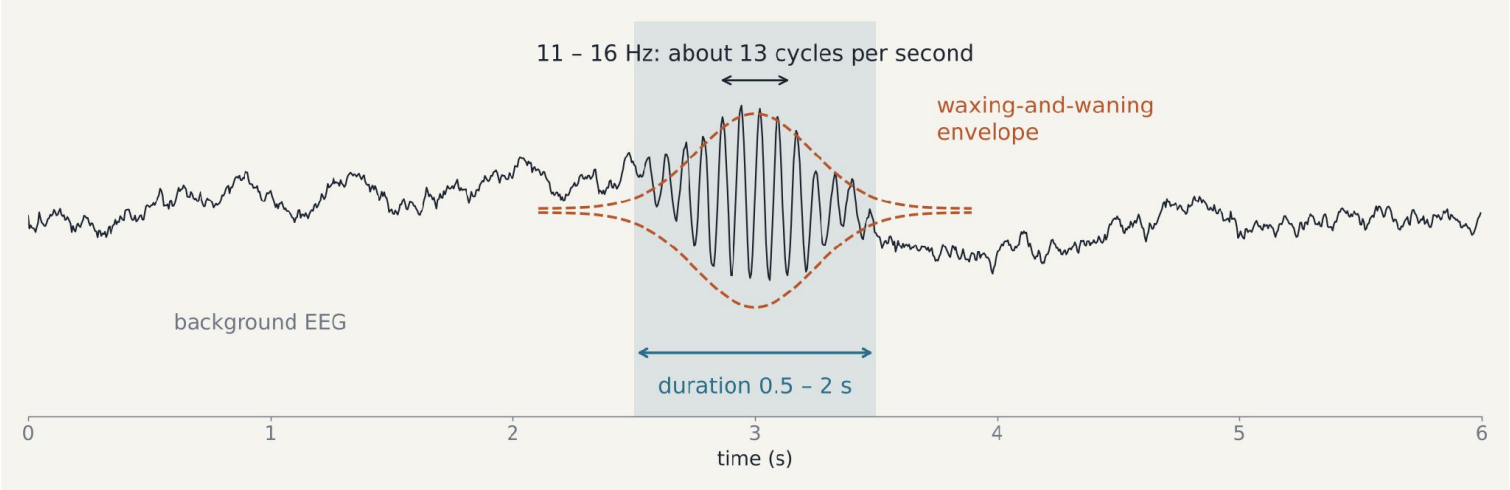
Benchmarking on Public Datasets and Transfer to in house Custom Dataset

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Sleep Spindle: A short burst of fast oscillation that grows and fades



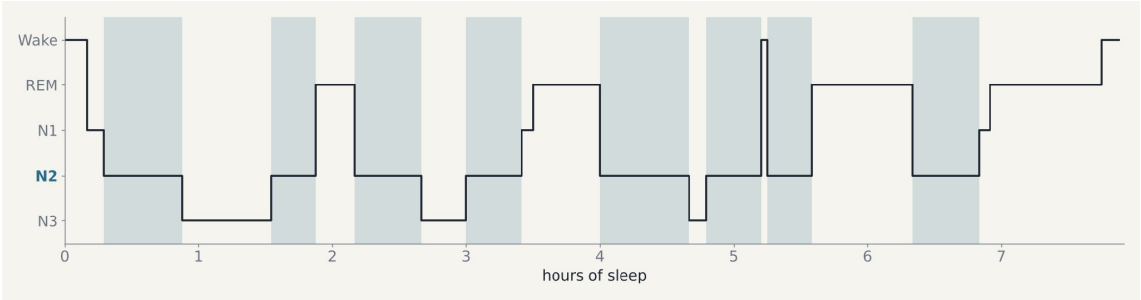
Shape
11–16 Hz, 0.5–2 s, rising and falling amplitude

Origin
Generated by thalamocortical loops; strongest over central electrodes

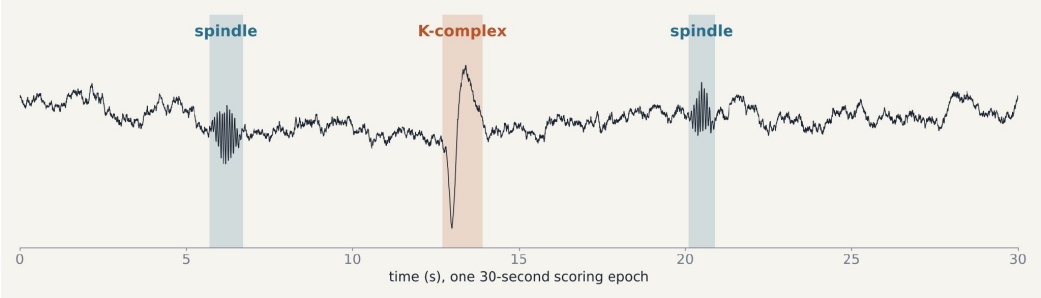
Why it matters
Linked to memory consolidation; density changes with age and disease

Spindles and K-complexes are what defines stage N2

A night of sleep, scored in 30-second epochs. N2 (shaded) is about half of it.



Zoom into one N2 epoch: what the technician looks for to call it N2.



Starting with two datasets and one detector

The proposal already lists the datasets and detectors the benchmark needs.

Datasets

DREAMS: loaded. Short excerpts, spindles marked by two experts.

NAP EEG: 64 channels, spindles, K-complexes and sleep stages.

Detector

YASA: the classic rule-based detector. Filters to the spindle band and applies thresholds.

It is the natural baseline: every learned detector should be compared against it.

DREAMS

Origin

- Université de Mons, Belgium (Devuyst et al.)
- Published on Zenodo; free download
- Licence CC BY-NC-ND: free to use, no redistribution of modified versions
- Also contains K-complex, REM, apnea and whole-night subsets; spindles and K-complexes are scored for events

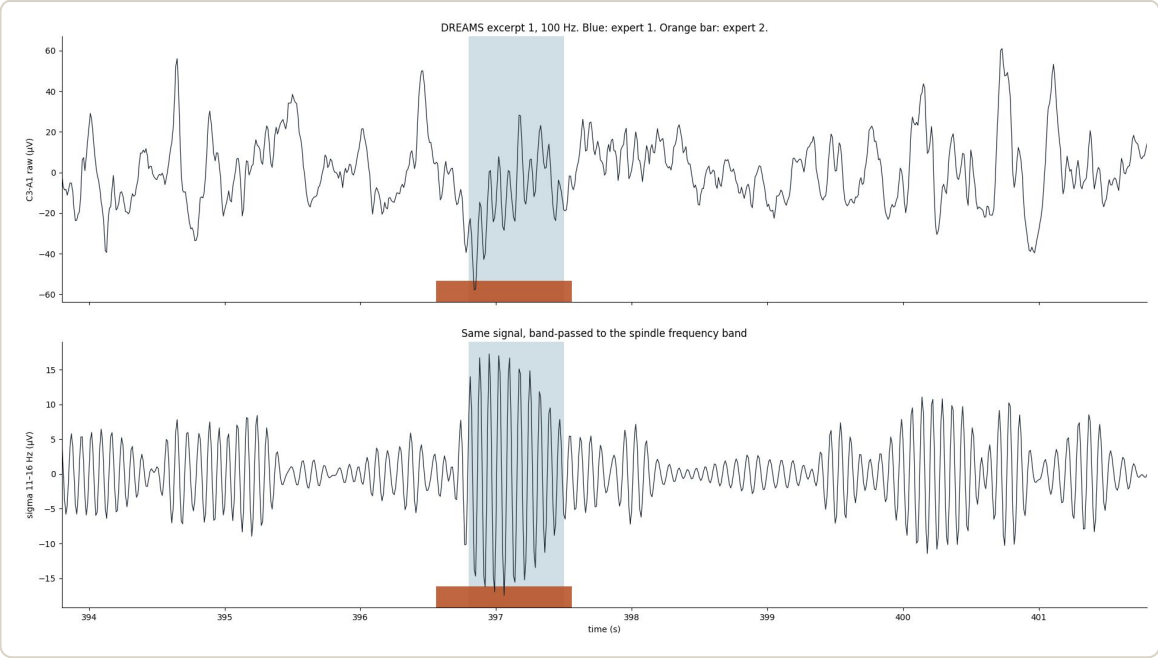
Recordings

- 8 excerpts, 30 minutes each, one per subject: 4 hours in total
- Two setups: excerpts 1 and 3 have 20 channels, scored on C3-A1; the other six have 28, scored on Cz-A1
- Sampling rate: 200 Hz for six, 100 Hz in excerpt 1, 50 Hz in excerpt 3
- Clinical recordings: oxygen, breathing and CPAP channels included

Annotations

- One text file per expert per excerpt: onset and duration in seconds
- Expert 1 scored all 8; expert 2 only 6
- Expert 2 gave every spindle a fixed 1-s duration
- Expert 1 stops scoring after about 1000 s in some excerpts
- Sleep stages in every file, one per 30 s: only 59% is N2; the rest N1, REM, N3, wake

DREAMS



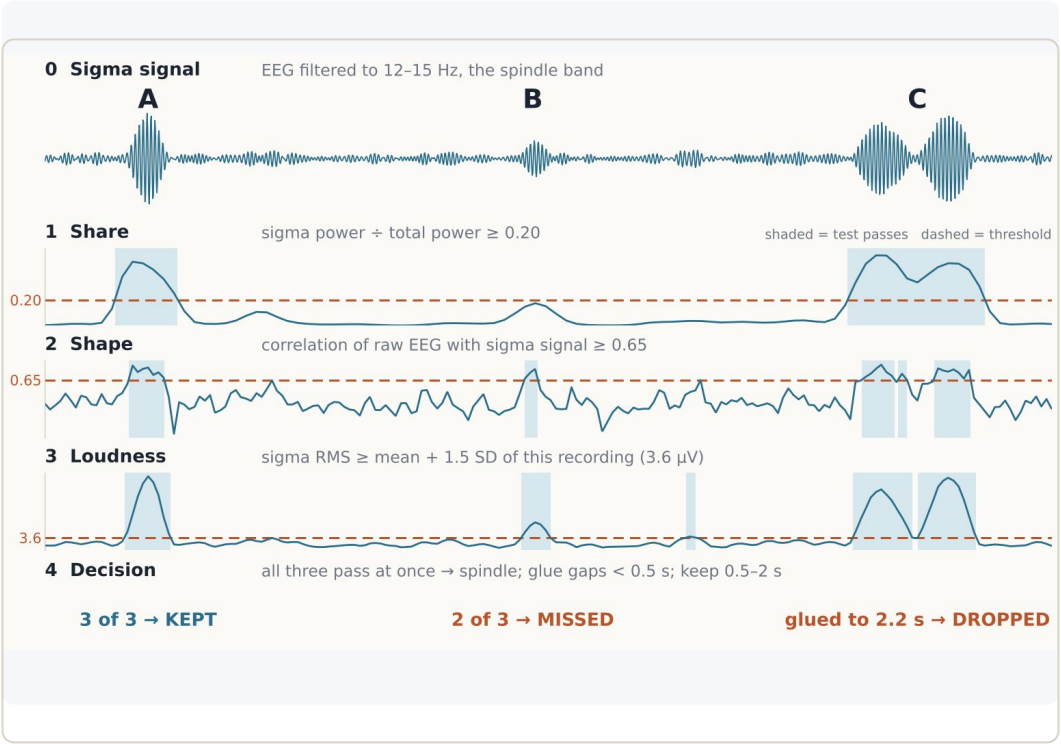
Excerpt 1, 30 min, 100 Hz, C3-A1

	Expert 1	Expert 2
spindles	52	115
durations	0.50–1.67 s	all 1.00 s
scored until	993 s	whole excerpt

Top: raw EEG. The spindle is the tight 13 Hz ripple inside the band, about 20 µV on a 40 µV background. Hard to see by eye, which is why scorers disagree.

Bottom: the same signal band-passed to 11–16 Hz. The spindle stands up as a burst. This trace is what every detector actually thresholds.

How YASA decides a spindle is there



1 · Relative sigma power

$$P_{\mathrm{rel}}(t) = \frac{\sum_{12\text{–}15\,\mathrm{Hz}} |X_b(t,f)|^2}{\sum_{1\text{–}30\,\mathrm{Hz}} |X_b(t,f)|^2} \;\geq\; 0.20$$

STFT of xb, 2-s window, 0.2-s step

2 · Moving correlation

$$r(t) = \mathrm{corr}\,(x_\sigma,\; x_b)_{\,t\,\pm\,0.15\,\mathrm{s}} \;\geq\; 0.65$$

Pearson, 0.3-s window, 0.1-s step

3 · Moving RMS

$$\mathrm{RMS}(t) = \sqrt{\mathrm{mean}\,(x_\sigma^2)}_{\,t\,\pm\,0.15\,\mathrm{s}} \;\geq\; \mu + 1.5\,\sigma_{\mathrm{trim}}$$

μ, σ from this recording, 10% trimmed; capped at 10 µV

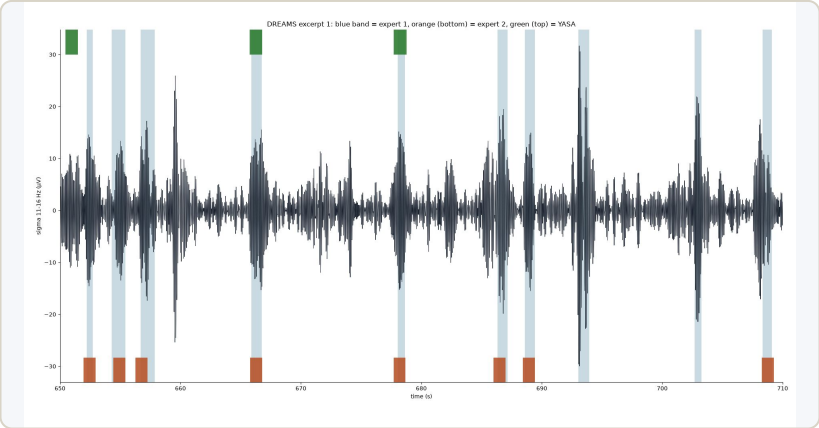
4 · Decision

$$n(t) = \mathbf{1}_1 + \mathbf{1}_2 + \mathbf{1}_3, \qquad \text{spindle} \;\Leftrightarrow\; \bar{n}_{100\,\mathrm{ms}}(t) > 2$$

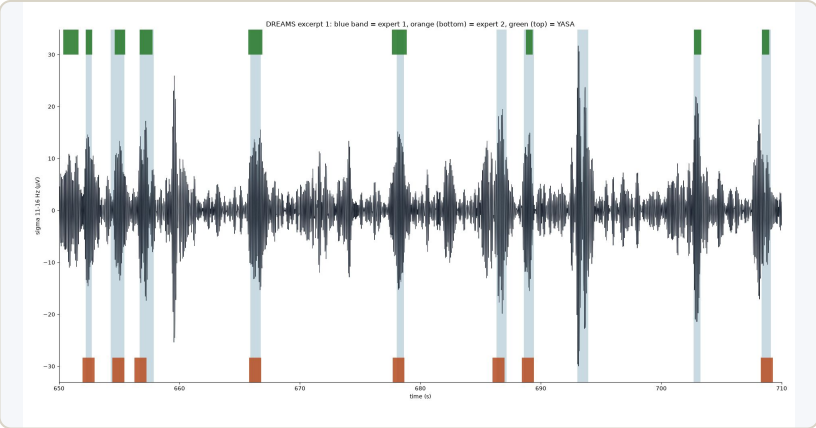
needs all three; then merge gaps < 0.5 s, keep 0.5–2 s

YASA on DREAMS

Published defaults: 3 detections



Thresholds loosened: 9 detections



60-s window, expert 1 marked 10	Defaults	Loosened
YASA detections	3	9
hits on expert-1 spindles	2	7
false positives	1	1, the same

Green = YASA, blue = expert 1, orange = expert 2. Loosening three thresholds triples recall at no cost in precision: RQ4's cheapest rung, shown necessary before any learned model.

NEXT STEPS

A formal YASA pipeline on DREAMS, then NAP EEG

DREAMS

1 · Formalise the current pipeline

- Run YASA on all 8 excerpts, not one window
- Formal criteria: one-to-one matching by onset, precision, recall and F1 per excerpt
- Scored against expert 1, expert 2 or decide what to do here
- Whole excerpt and N2 only, using the stage channel

NAP EEG · ABOUT 14 GB

2 · Look into the dataset

Download and open it, as with DREAMS: files, channels, sampling rate, annotations

3 · Analyse NAP EEG

Spindle counts, scorer agreement, sleep stages

4 · YASA detection pipeline on NAP EEG

Same criteria as DREAMS, so the results are comparable